Chapter 3

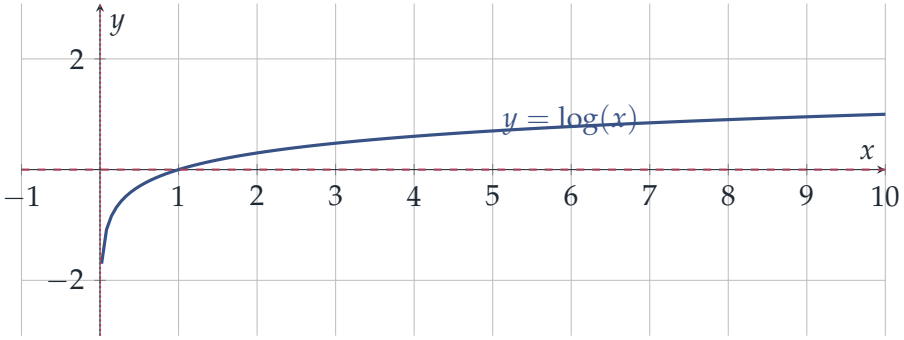
Common logarithm function

Skills: switch from exponential to logarithmic form, solve, model

3.1 Reference notes

Key ideas

For every $b > 0$, $\log(b)$ is the unique real number x such that $10^x = b$. Thus $\log(10^x) = x$ and, for $b > 0$, $10^{\log(b)} = b$. The log function is defined on $]0, +\infty[$ and it is strictly increasing.



For $a > 0$, $b > 0$ and every natural number n :

$$\log(ab) = \log(a) + \log(b), \qquad \log\left(\frac{a}{b}\right) = \log(a) - \log(b), \qquad \log(a^n) = n\log(a).$$

Common mistake

$\log(a+b)$ does not simplify. Before any logarithmic calculation, check that each argument is strictly positive.

Method

To solve $a^x = b$, with $a > 0$, $a \neq 1$ and $b > 0$, we take the logarithm: $x\log(a) = \log(b)$, therefore $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Solve $1{,}08^x = 2$ amounts to writing $x = \dfrac{\log(2)}{\log(1{,}08)} \approx 9{,}01$. A value multiplied each year by 1.08 therefore doubles after a little more than nine years.

Since $\log$ is increasing, for $a, b > 0$, $a < b$ is equivalent to $\log(a) < \log(b)$. For $a > 1$, $a^x < b$ amounts to $x < \dfrac{\log b}{\log a}$; when $0 < a < 1$, the inequality is reversed.

Session 15 — Definition of the common logarithm

Common logarithm function

17/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Link powers of 10 and logarithms.

· Calculate standard values.

· Use the calculator.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 20 per cent of 80 .....
2. Give the domain of log
3. Calculate $\log(10^4)$. .....
4. Solve $10^x = 0{,}01$. .....
5. Expand $\log(3b)$, $b > 0$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: link powers of 10 and logarithms and calculate standard values.

Investigation

Complete the pairs $10^x = b$ and $x = \log(b)$ for $b = 0{,}001; 0{,}1; 1; 10; 1{,}000$. Then use the calculator for $b = 7$.

Written summary

For $b > 0$, $\log(b)$ is the unique solution of $10^x = b$. In particular $\log(1) = 0$, $\log(10) = 1$ and $\log(100) = 2$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate $\log(10^2)$ and $\log(10^{-2})$.
2. Expand $\log\left(\frac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $2^x = 22$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao pp. 18-21; ex. p. 24 #1-3. The exercise in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 70, Exercise 2

Solve the following equations.

a. $5 \times 10^{x} = 3{,}375$

b. $3{,}2 + 2 \times 10^{x} = 4{,}5 \times 10^{x}$

c. $-17{,}3 + 10^{x} = 5 - 3 \times 10^{x}$

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 195 and the rate or common ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 16 — Direction of variation and representation

Common logarithm function

19/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Know the domain.

· Use monotonicity.

· Read the curve.

Fluency · 6 to 8 minutes, without a calculator

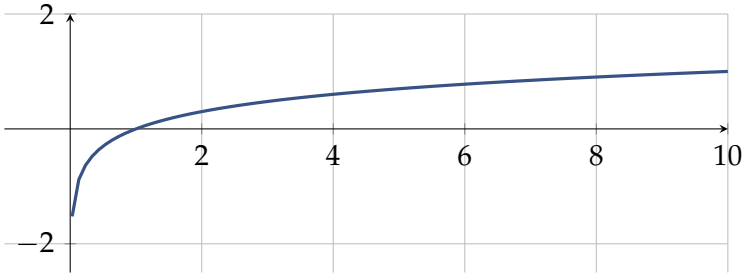
1. Write the multiplier for a 25% increase
2. Calculate $\log(10^0)$. .....
3. Solve $10^x = 0{,}01$. .....
4. Expand $\log(3b)$, $b > 0$. .....
5. Give the domain of $\log$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: know the domain and use monotonicity.

Investigation

Draw $y = 10^x$ and the line $y = x$, then obtain the graph of the logarithm by symmetry. Deduce its domain, its direction of variation and the asymptote.



Written summary

The function $\log$ is defined on $]0, +\infty[$, strictly increasing, and its graph has $x = 0$ as a vertical asymptote.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate $\log\left(10^{3}\right)$ and $\log(10^{-3})$.
2. Expand $\log\left(\dfrac{100a^{2}}{b}\right)$ for $a, b > 0$.
3. Solve $3^{x} = 23$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao pp. 18-21; ex. p. 24 #1-3. The exercise in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 70, Exercise 7

Give the sign of the following numbers.

a. $\log(2{,}5)$ b. $\log(0{,}25)$ c. $\log\left(\dfrac{7}{10}\right)$

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 200 and the rate or common ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 17 — Algebraic properties

Common logarithm function

24/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Expand or condense logarithms.

· Check the conditions.

· Avoid false properties.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 6 - x + 2$. .....
2. Solve 10 $^{x} = 0{,}01$. .....
3. Expand log $(3b)$, $b > 0$. .....
4. Give the domain of log
5. Calculate log (10^{1}). .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: expand or combine logarithms and check the conditions.

Investigation

From 10 $^{2} \times 10^{3} = 10^{5}$, conjecture $\log(ab) = \log a + \log b$. Test numerically with $a = 2, b = 5$, then explain why a, b must be positive.

Written summary

For $a, b > 0$, $\log(ab) = \log a + \log b$ and $\log\left(\frac{a}{b}\right) = \log a - \log b$. Moreover log $(a^{n}) = n \log a$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate log (10^{4}) and $\log(10^{-4})$.
2. Expand log $\left(\frac{100a^{2}}{b}\right)$ for $a, b > 0$.
3. Solve 4 $^{x} = 24$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao pp. 18-21; ex. p. 24 #1-3. The exercise in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 73, Exercise 24

Write in the form $\log(A)$ for A is a real number to be specified, the following numbers:

a. $\log(2)+\log(7)-\log(5)$

b. $\log(3)-2\log(5)$

c. $\log(3)+\log(7)$

d. $3\log(7)-7\log(3)$

e. $\log(12)-\log(4)+2\log(3)$

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 205 and the rate or common ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 18

Common logarithm function

Solve $10^x = b$

26/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve $10^x = b$.

· Bound a solution.

· Interpret an order of magnitude.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 5 per cent of 140 .....
2. Expand $\log(3b)$, $b > 0$. .....
3. Give the domain of $\log$
4. Calculate $\log(10^2)$. .....
5. Solve $10^x = 0{,}01$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: solve $10^x = b$ and bracket a solution.

Investigation

A device measures intensities $b = 0{,}04; 3{,}2; 850$. For each value, solve $10^x = b$, give x to the nearest hundredth and check by calculating 10^x.

Written summary

The equation $10^x = b$ has a solution only if $b > 0$; this solution is $x = \log(b)$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate $\log(10^5)$ and $\log(10^{-5})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $5^x = 25$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao, p. 73, No. 30 to 32. The exercise reproduced in this session is identified below. by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 73, Exercise 32

Harpagon invested a sum of 24,000 €, with compound interest at an annual rate of 3,5%. A bit later, he decides to close this account when the balance is 40 208 ,37 €. The aim of the exercise is to determine the number of years the account has existed.

1. Complete the unfinished algorithm so that it answers the problem.
2. Determine how many years the account has existed.
3. Translate the algorithm into Python and enter it in a Python editor. Compare the result obtained with that of question 2.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 210 and the rate or common ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 19

Solve $a^x = b$

Common logarithm function

01/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Solve $a^x = b$.

· Use a logarithm quotient.

· Round sensibly.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 10% increase
2. Give the domain of log
3. Calculate log (10^3). .....
4. Solve $10^x = 0{,}01$. .....
5. Expand log $(3b)$, $b > 0$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: solve $a^x = b$ and use a quotient of logarithms.

Investigation

The number of subscribers is multiplied by 1,15 each month. Determine after how many months it will have tripled. Set up the equation, solve it using logarithms and interpret the rounding.

Written summary

For $a > 0$, $a \neq 1$ and $b > 0$, $a^x = b$ is equivalent to $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate log (10^6) and $\log(10^{-6})$.
2. Expand log $\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $6^x = 26$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao, p. 74, Nos. 33 to 39. The exercise reproduced in this session is identified below. by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 73, Exercise 30

1. A first capital of 6,000 € is invested at compound interest at a rate of 9%. After how many years will this capital have doubled? Tripled?

2. A capital of 9,000 € is invested on the same day at a rate of 6%. After how many years will the value of the first capital have exceeded that of the second?

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 215 and the rate or common ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 20 — Equations $x^a = b$ and simple inequalities

Common logarithm function

03/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve for an unknown exponent.

· Handle an inequality.

· Take the domain into account.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 9 - x + 2$. .....
2. Calculate log (10^4). .....
3. Solve 10 $^x = 0{,}01$. .....
4. Expand log $(3b)$, $b > 0$. .....
5. Give the domain of log

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: solve for an unknown exponent and handle an inequality.

Investigation

Solve on $[0, +\infty[$: $x^3 = 40$, then $x^4 = 16$. Then solve 1 $,08^x < 1{,}5$ and explain the role of monotonicity.

Written summary

To solve $x^n = b$, take into account the domain and the parity of n. For an exponential inequality, the direction depends on whether the function is increasing or decreasing.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate log (10^2) and $\log(10^{-2})$.
2. Expand log $\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve 2 $^x = 22$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao, p. 75, nos. 40 to 42. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 74, Exercise 36

A capital of 15,500 € is invested at compound interest for four and a half years. A second capital of 16 480 € is also invested at compound interest for 5 years and 3 months at an annual rate of 6%. What must the interest rate on the first capital be so that the amounts at the end of the investment are identical?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 220 and the rate or common ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 21 — Problems in context

Common logarithm function

08/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Model a situation.

· Choose an equation.

· Conclude in context.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 20 per cent of 60 .....
2. Solve $10^{x} = 0{,}01$. .....
3. Expand $\log(3b)$, $b > 0$. .....
4. Give the domain of $\log$
5. Calculate $\log(10^{3})$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: model a situation and choose an equation.

Investigation

The sound level is $L = 10\log\left(\frac{I}{I_0}\right)$. Calculate L when $I = 10^{6} I_0$. Then determine the ratio $\frac{I}{I_0}$ for $L = 85$ dB.

Written summary

In a problem, define the variable, set up the equation, solve, check the domain and conclude with the unit. A numerical solution without interpretation is incomplete.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate $\log(10^{3})$ and $\log(10^{-3})$.
2. Expand $\log\left(\frac{100a^{2}}{b}\right)$ for $a, b > 0$.
3. Solve $3^{x} = 23$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao, p. 74, No. 38·39, then pp. 75, No. 40 to 42. The exercise reproduced in this sitting is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 75, Exercise 41

The dynamic range of an image is the ability of a digital image to render the scales of light intensities in the real world. Dynamic range is often expressed in stops. In digital imaging, the darkest pixel (black) corresponds to 0 and the brightest pixel (white) corresponds to 255. There are thus 256 light intensities, which in computing terms translates into 8 bits of precision ($256 = 2^8$). The number of stops is: $\frac{\log(256)}{\log(2)} = 8$.

1. For an underexposed digital photo, only 128 light intensities can be captured. How many stops does this correspond to?
2. A RAW file has 10-bit precision. What is the corresponding number of stops?
3. A standard video camera has a dynamic range of 5 ,5 stops. How many different light intensities does it capture?
4. The latest-generation television screens have a dynamic range of 16 stops. How many different light intensities can they display?
5. The number of light intensities in the real world is of the order of 100,000,000. How many stops does this correspond to?

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 225 and the rate or common ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 22 — Review and assessment

Common logarithm function

10/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Summarise the methods.

· Identify the conditions.

· Fix errors.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 25% increase
2. Expand $\log(3b)$, $b > 0$. .....
3. Give the domain of $\log$
4. Calculate $\log(10^4)$. .....
5. Solve $10^x = 0{,}01$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: summarise the methods and identify the conditions.

Investigation

Make a method map: domain; properties; equations $10^x = b$; equations $a^x = b$; inequalities. Match each branch with a condition and an example.

Written summary

Before computing a logarithm: check positivity. Before a comparison: identify the direction of variation. Before concluding: check plausibility.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $4^x = 24$ and give an approximation to the nearest hundredth.

References in the textbook

Extensions: Calao, pp. 76-79. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources

digital resources of the textbook.

Calao textbook · p. 75, Exercise 42

N is a non-zero natural number. The aim of the exercise is to find a method to determine the number of digits in its decimal representation.

1. $N = 10\,203$.
   a. Bound N between two consecutive powers of 10. Deduce bounds for $\log(N)$ between two consecutive integers.
   b. Deduce the value, rounded up to the nearest integer, of $\log(N)$. Compare this result with the number of digits of N.
2. N has 23 digits.
   a. Bound N between two consecutive powers of 10. Deduce bounds for $\log(N)$ between two consecutive integers.
   b. Give the value, rounded up to the nearest integer, of $\log(N)$. Which value do we recover ?
3. Deduce from the previous questions a method for finding the number of digits of each non-zero natural number when it is given in decimal form.
4. Use the previous method to find the number of digits of the following integers: a. 7^{49} ; b. 5658 ; c. $2\,019 \times 10^{20}$.
5. The largest prime number known to date is a Mersenne number, which is written: $2^{82\,589\,933} - 1$. How many digits does it have?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 230 and the rate or common ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.